

CNSA172 Web Technologies and Network Security

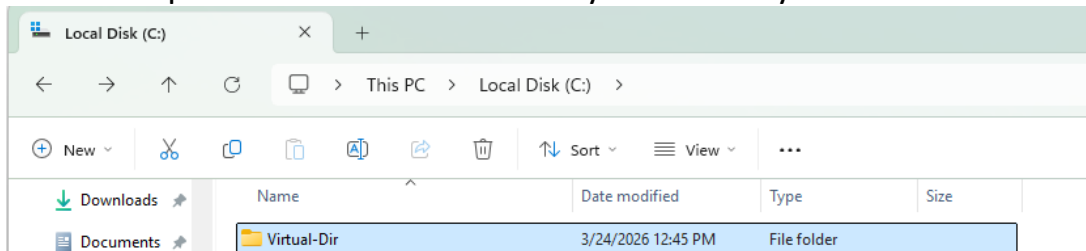
Virtual Web Server Mechanisms Lab

Goal: Create virtual directories and multiple websites on your IIS server. Take your time with this lab; be sure to absorb its contents.

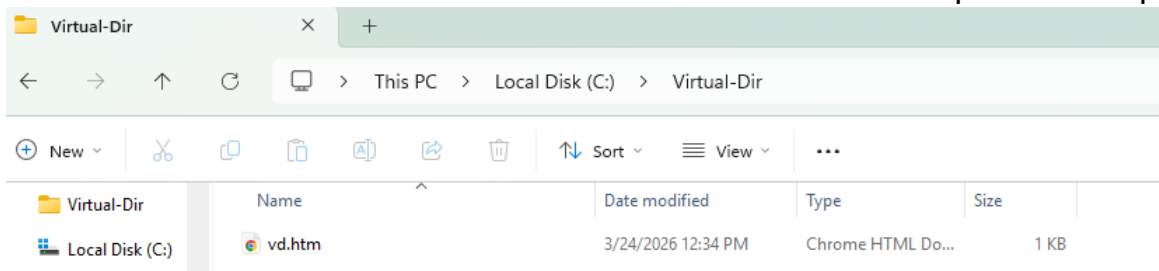
Materials: Windows 11 with IIS installed

VIRTUAL DIRECTORIES:

1. Use File Explorer to create a folder on your PC's %systemdrive% drive named **Virtual-Dir**.



2. Use the editor of your choice to create a simple HTML document named **vd.htm** and place the document in the **Virtual-Dir** folder that was created in the previous step.



3. Use IIS Manager to create a virtual directory with an alias of **viridir** and associate it with the directory named **Virtual-Dir**.

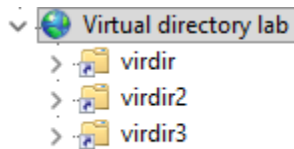
Right click on a site and click add virtual directory, set alias to viridir, and put the physical path as C:\Virtual-dir

4. At this point, your Web server has a new virtual directory named **viridir** that contains one document named **vd.htm**. To use a Web browser to view this document, you will have to include the name of the virtual directory in your Web Browser's address bar. You will need to have the properties of the virtual directory set correctly to serve the document. Use your Web Browser and your knowledge of IIS Administration to successfully access the contents of the created virtual directory. **Record the exact web address you used to access this document here. Also, note any other IIS adjustments you discovered that were necessary to successfully serve this document to a client's web browser.**

I had to put C:/Virtual-Dir/vd.htm, to access it, I didn't have to change anything in IIS

5. Repeat the previous steps so that ultimately your website has 3 virtual directories. Name the second virtual directory **viridir2**, and the third one **viridir3**. Each virtual directory

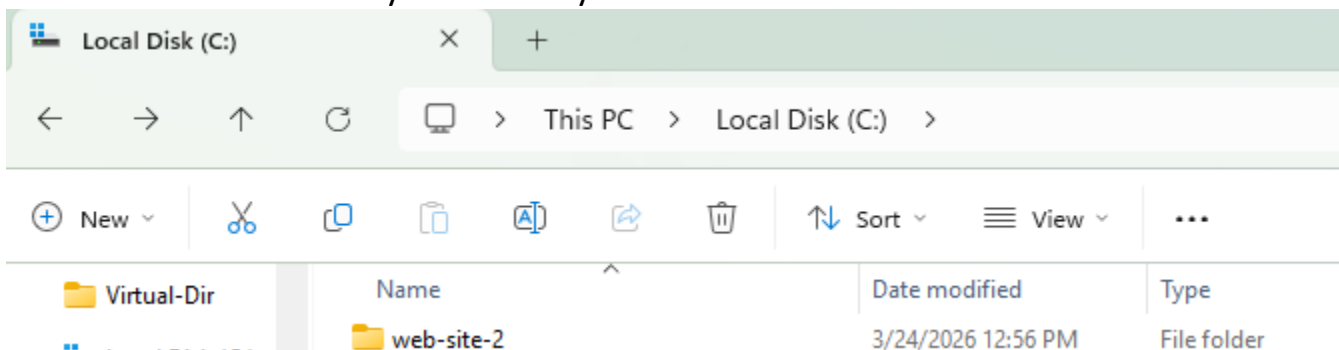
should contain a different document of your making that clearly allows the client to confirm the difference between documents in one virtual directory and those in another. Be sure to test each virtual directory to ensure it is fully operational. **Insert a snip of your virtual directories and their location in the directory structure:**



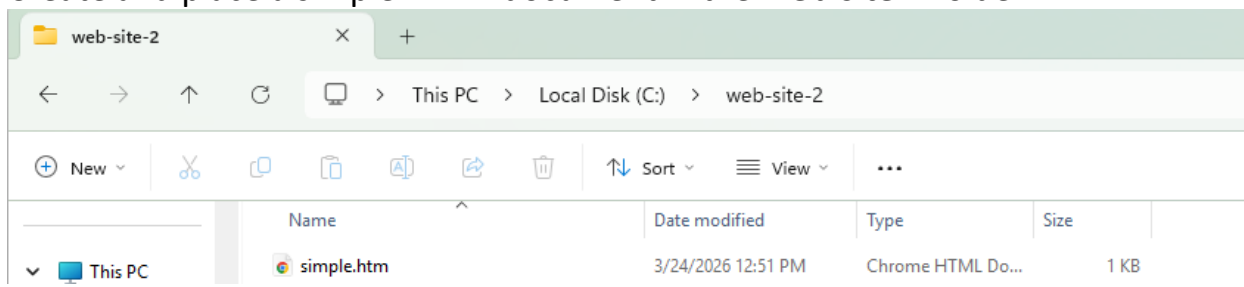
VIRTUAL SITES (MULTIPLE WEB SITES ON A SINGLE SERVER):

For this part, you will have to work and think carefully. Small details are purposefully left out to provoke some thought and troubleshooting skills.

1. Create a folder on your PC's %systemdrive% drive called **web-site-2**.



2. Create and place a simple HTML document in the **web-site-2** folder.



3. Create a new website on your web server called **site2** (Use **site2** when prompted for the description). Keep in mind that to have one computer host two different websites, you'll want to use "Host Headers" and address the other considerations that go into making a website accessible. See next step.

Right click on the sites folder, click add website, put the site name as site2, put the physical path on C:\site2, set the host name to www.site2.com, and leave it as port 80

4. Use a host header name of www.site2.com and indicate that the home directory will be **web-site-2**. Set security for this new site to anonymous access only (remove Windows authentication). Be sure to set the default document name correctly.
Go to authentication and turn on anonymous authentication, go to default document and put the top option as simple.htm
5. Enter www.site2.com into your browser. You might find that the results you get are different from what you expected. **Explain what you think happened and why you think it happened: Google choose to go to a website registered online with the url www.site2.com as it defaults to that as opposed to the html site I just made.**
6. Try this - Can you fix the situation that occurred above using the hosts file located on your PC in the \windows\system32\drivers\etc folder? Keep in mind the hosts file is just a potential workaround solution to provide access to your second website on your PC only. In the real world, DNS records would have to be added to the DNS servers that your clients look to for name resolution. **Insert a snip of your modified host file here:**

```
# Copyright (c) 1993-2009 Microsoft Corp.
#
# This is a sample HOSTS file used by Microsoft TCP/IP for Windows.
#
# This file contains the mappings of IP addresses to host names. Each
# entry should be kept on an individual line. The IP address should
# be placed in the first column followed by the corresponding host name.
# The IP address and the host name should be separated by at least one
# space.
#
# Additionally, comments (such as these) may be inserted on individual
# lines or following the machine name denoted by a '#' symbol.
#
# For example:
#
#       102.54.94.97       rhino.acme.com          # source server
#       38.25.63.10       x.acme.com             # x client host


# localhost name resolution is handled within DNS itself.
#       127.0.0.1         localhost
#       ::1               localhost
127.0.0.1 www.site2.com
|
```

7. Ultimately, you should be able to reach a point where you can type in www.site2.com into your browser's address box and see the web page you created come up. Be sure that this works and that you understand how before proceeding any further.
8. Modify site2's properties so that it uses TCP Port 5200. Then try using your browser to connect to www.site2.com. **What happens?**
It can no longer connect to the site and instead goes to my default port 80 website

9. How do you connect to a website (your web site2) that does not use the standard HTTP port 80? **Insert a snip of your browser and include the URL you typed along with your unique html page:**

